

Summary

Database management systems (DBMS) expert with strong systems fundamentals spanning query engines, storage architectures, and operating systems (OS). Deep experience in analytical and transactional DBMS design: query execution and join algorithms, adaptive query execution and dynamic data pruning, storage management, and concurrency control. Complementary strengths in distributed systems, open table and file formats, and low-level performance engineering at the OS/DBMS boundary (CPU scheduling, eBPF, kernel networking, and observability). Active in the research and open source communities as a PMC member of Apache DataFusion and a frequent program committee member and reviewer, including for ACM SIGMOD and VLDB.

Professional Experience

2024 –	Apple Senior Distributed Database Engineer	Washington, DC
	- Enhancing Apple's data lakehouse by contributing to multiple open source Apache projects: Arrow, DataFusion, DataFusion Comet, Iceberg, Parquet, and Spark. - Productionized Comet, demonstrating >50% cost reduction on customer Spark jobs. - Enabled Comet's dynamic partition pruning (adaptive and non-adaptive query execution), which previously forced queries to fall back to slower Spark execution. Across 78 TPC-DS benchmark queries, the share of each query running natively rose from ~30–50% to ~95–97%, contributing to a >50% overall speedup at 1 TB. - Designed and shipped Comet's native Iceberg scan, upstreaming >30 PRs to Iceberg. - Drove a DataFusion sort-merge-join (SMJ) overhaul across ~12 PRs, making TPC-H >500x faster and near-unique LEFT/FULL joins >20x faster. - Diagnosed and fixed a Comet busy-poll regression, dropping cluster CPU utilization from 30% to 8% on I/O-bound Iceberg workloads. - Re-architected the JVM-to-native data path to true zero-copy using the Arrow C Stream Interface, eliminating a per-batch deep copy on every batch crossing the boundary.	
2009 – 2017	Western Digital Corporation Lead Technician V	Irvine, CA
	- Led a team of 17 technicians verifying firmware for HDD and SSD storage devices. - Promoted four times from initial role as Technician II while earning my B.Sc.	

Education

2019 – 2024	Carnegie Mellon University Doctor of Philosophy (Ph.D.), Computer Science	Pittsburgh, PA
	- Advisor: Andy Pavlo - Thesis: "On Embedding Database Management System Logic in Operating Systems via Restricted Programming Environments"	
2017 – 2018	Carnegie Mellon University Master of Science (M.Sc.), Computer Science	Pittsburgh, PA
2013 – 2017	University of California, Irvine Bachelor of Science (B.Sc.), <i>cum laude</i> , Computer Science and Engineering	Irvine, CA
2010 – 2013	Irvine Valley College Certificate of Achievement, Computer Science: Computer Languages	Irvine, CA

Selected Talks

2025	CS 224P: Big Data Systems “Fusing Academic Research and Open Source Systems for Petabyte-Scale Analytics”	Link YouTube
2025	DC Systems “Putting Theory Into Practice”	Link YouTube
2023	15-445/645: Database Systems Lecture on join algorithms	Link YouTube
2019	15-445/645: Database Systems Lecture on two-phase locking concurrency control	Link YouTube

Selected Publications

2025	Matthew Butrovich , Samuel Arch, Wan Shen Lim, William Zhang, Jignesh M. Patel, and Andrew Pavlo. “BPF-DB: A Kernel-Embedded Transactional Database Management System For eBPF Applications”. <i>SIGMOD Conference</i> .
2023	Matthew Butrovich , Karthik Ramanathan, John Rollinson, Wan Shen Lim, William Zhang, Justine Sherry, and Andrew Pavlo. “Tigger: A Database Proxy That Bounces With User-Bypass”. <i>Proc. VLDB Endow.</i>
2022	Matthew Butrovich , Wan Shen Lim, Lin Ma, John Rollinson, William Zhang, Yu Xia, and Andrew Pavlo. “Tastes Great! Less Filling! High Performance and Accurate Training Data Collection for Self-Driving Database Management Systems”. <i>SIGMOD Conference</i> .

Service

To the Open Source Community

2026 –	PMC Member, Apache DataFusion, <i>The Apache Software Foundation</i> .	Link
2025 –	Contributor, Apache Arrow, <i>The Apache Software Foundation</i> .	Link
2025 –	Contributor, Apache Iceberg, <i>The Apache Software Foundation</i> .	Link
2025 –	Contributor, Apache Parquet, <i>The Apache Software Foundation</i> .	Link
2024 –	Contributor, Apache Spark, <i>The Apache Software Foundation</i> .	Link
2025 – 2026	Committer, Apache DataFusion, <i>The Apache Software Foundation</i> .	Link
2024 – 2025	Contributor, Apache DataFusion, <i>The Apache Software Foundation</i> .	Link

To the Profession

2026	Member, <i>ACM SIGMOD 2027 Program Committee</i> .	Link
2026	Member, <i>VLDB 2027 Program Committee</i> .	Link
2026	Member, <i>4th Workshop on eBPF and Kernel Extensions Program Committee</i> .	Link
2025	Member, <i>ACM SIGMOD 2026 Industry Track Program Committee</i> .	Link
2025	Member, <i>ACM SIGMOD 2026 Program Committee</i> .	Link
2025	Reviewer, <i>ACM Transactions on Database Systems (TODS)</i> .	Link
2024	Member, <i>eBPF Summit 2024 Program Committee</i> .	Link
2023	Member, <i>eBPF Summit 2023 Program Committee</i> .	Link
2023	Member, <i>ACM SIGMOD 2023 Artifacts & Reproducibility Committee</i> .	Link

Skills

Programming Languages: C, C++, eBPF, Java, Python, Rust, Scala

Development Tools: bpftrace, GDB/LLDB, Git, Intel VTune, perf